

Linear Regression Models

From One Predictor to Multivariable Prediction

Dr. Marcelo Feighelstein

What Is Regression?

Regression is **supervised learning** where the target variable is a **continuous number** — not a class label.

🏷️ Classification

Predicts a **category**
Examples: spam / not spam, disease / healthy, fraud / legitimate

📈 Regression

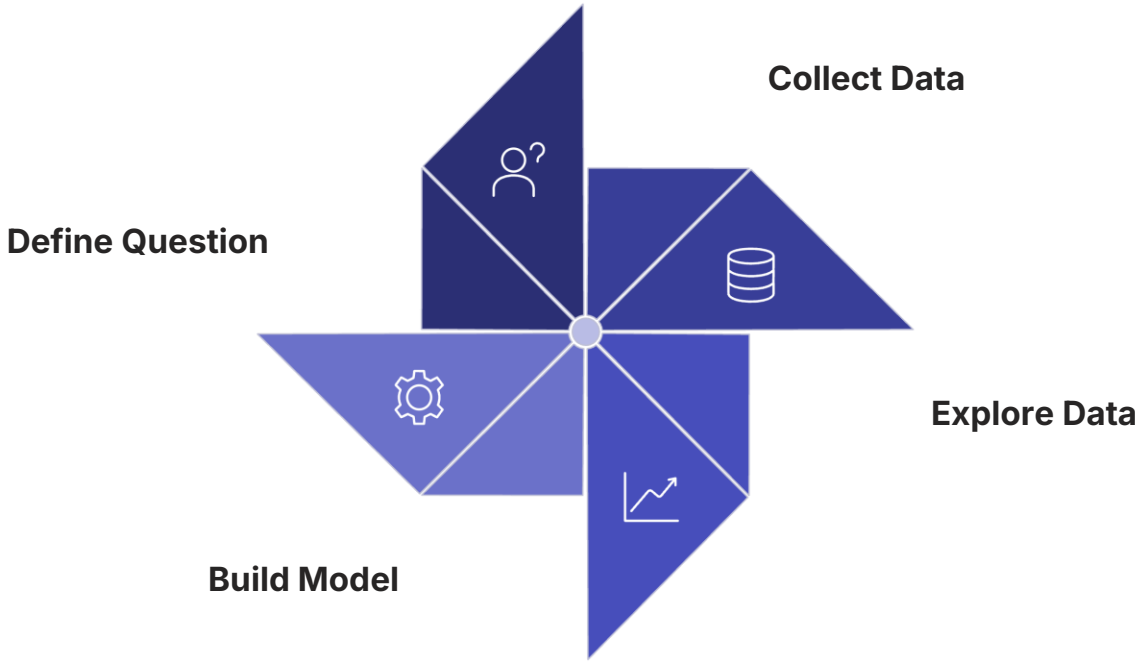
Predicts a **number**
Examples: house price, sales revenue, temperature, income, demand

ⓘ **Key question regression answers:** "How much?" or "What value will this be?"

Think of regression as drawing the best possible line (or surface) through your data to estimate values you haven't seen yet.

Regression in the Data Analysis Pipeline

Building a regression model is one step in a larger process. Understanding **where it fits** makes the whole workflow clearer.



Regression is not just a formula — it's embedded in a pipeline that starts with a real-world question and ends with an actionable insight. Skipping any step leads to poor or misleading models.

The California Housing Dataset

Our main dataset throughout this lecture: **median house prices** for California districts, drawn from 1990 census data.

Location

- longitude — how far west a house is; higher = farther west
- latitude — how far north a house is; higher = farther north
- oceanProximity — location of the house w.r.t ocean/sea

Housing

- housingMedianAge — median age of a house within a block
- totalRooms — total number of rooms within a block
- totalBedrooms — total number of bedrooms within a block

Demographics

- population — total number of people residing within a block
- households — total number of households within a block
- medianIncome — median income for households within a block of houses (measured in tens of thousands of US Dollars)

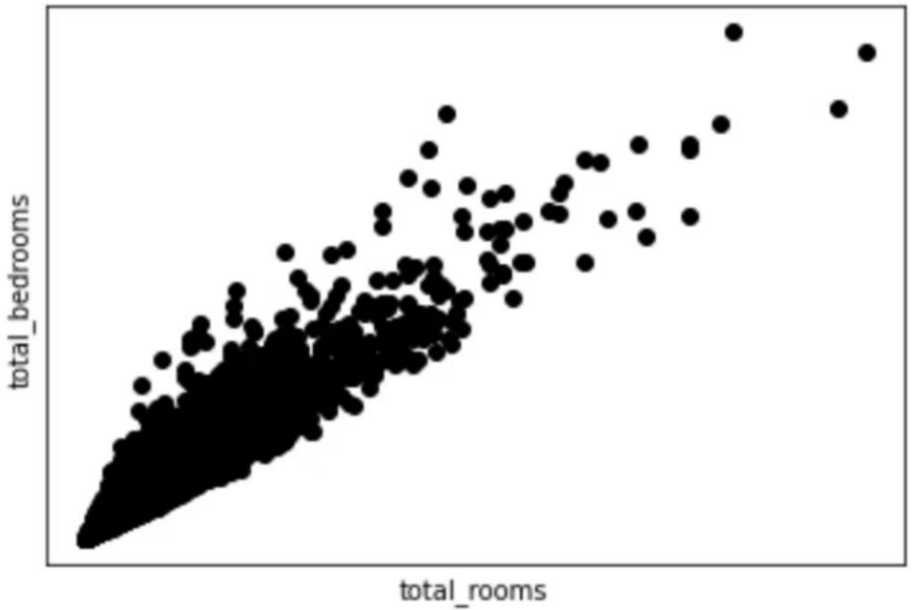
PREDICTION TARGET

Target

medianHouseValue — median house value for households within a block (measured in US Dollars)



Can You See a Linear Relationship?



Look at the scatter plot of total_rooms vs. total_bedrooms from the California Housing dataset.

> 💭 Do you think a linear model could capture this relationship? Why or why not?

- Does the data follow a straight-line trend?
- Are there any outliers or unusual patterns?
- Would a line through this data make good predictions?

Unpacking Linear Relationships: Key Questions

Before diving into predictive models, let's observe a simple relationship within our housing dataset: the number of total rooms often correlates directly with the number of total bedrooms. This linear trend sparks several important questions for our journey into regression analysis.

1 Calculating the Model

How do we mathematically determine the best-fit line's slope and intercept?

2 Evaluating Model Quality

What metrics and methods can we use to assess how well our model explains the data?

3 Assessing Linearity

How can we confirm if the underlying relationship in our data is genuinely linear?

4 Beyond One Predictor

What strategies do we employ when multiple variables influence the target, like housing price?

Calculating the Regression Line: β_0 and β_1

The values for the intercept (β_0) and slope (β_1) are not arbitrary. They are mathematically derived to define the unique line that best fits the observed data, specifically by minimizing the sum of the squared residuals.

1. Calculating the Slope (β_1)

$$\beta_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

- x_i, y_i: Individual data points
- $\bar{x}$, $\bar{y}$: The mean of the X and Y variables, respectively
- Σ: The sum of all values

This formula quantifies how Y changes with respect to X, considering the spread of both variables around their means.

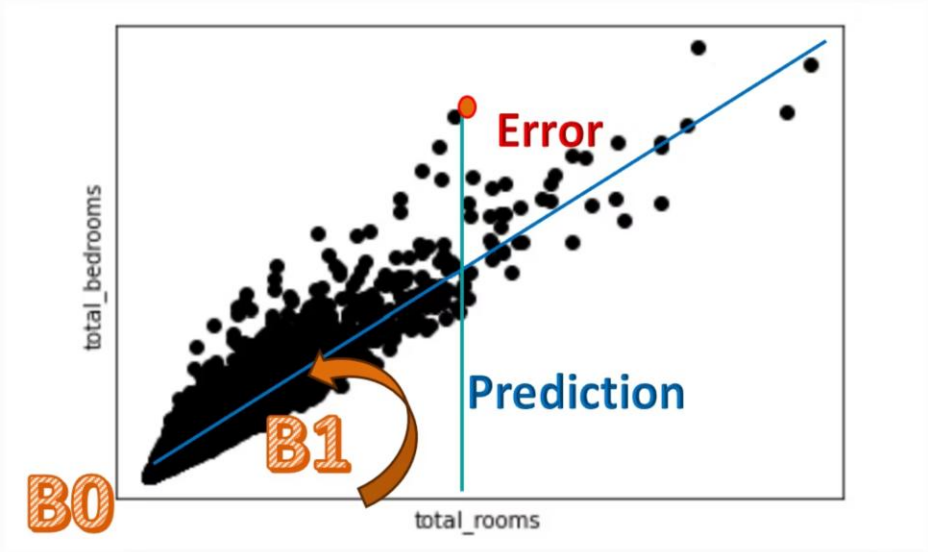
2. Calculating the Intercept (β_0)

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

- $\bar{y}$: The mean of the Y variable
- β_1: The calculated slope
- $\bar{x}$: The mean of the X variable

Once the slope (β_1) is determined, the intercept (β_0) can be found by ensuring the regression line passes through the point defined by the means of X and Y *(|bar{x}, |bar{y})*.

Simple Linear Regression: One Predictor



The simplest form: model the relationship between **one input X** and **one output Y**.

Formula: $\hat{Y} = \beta_0 + \beta_1 X$

- β_0 — Intercept: predicted Y when X = 0
- β_1 — Slope: change in Y for a one-unit increase in X

Housing example: predict *medianHouseValue* from *medianIncome*.

Understanding the Regression Line

The line gives a **predicted value** for every X. The gap between the real point and the line is the **prediction error**.



Some points fall above the line

The model under-predicted for these cases.

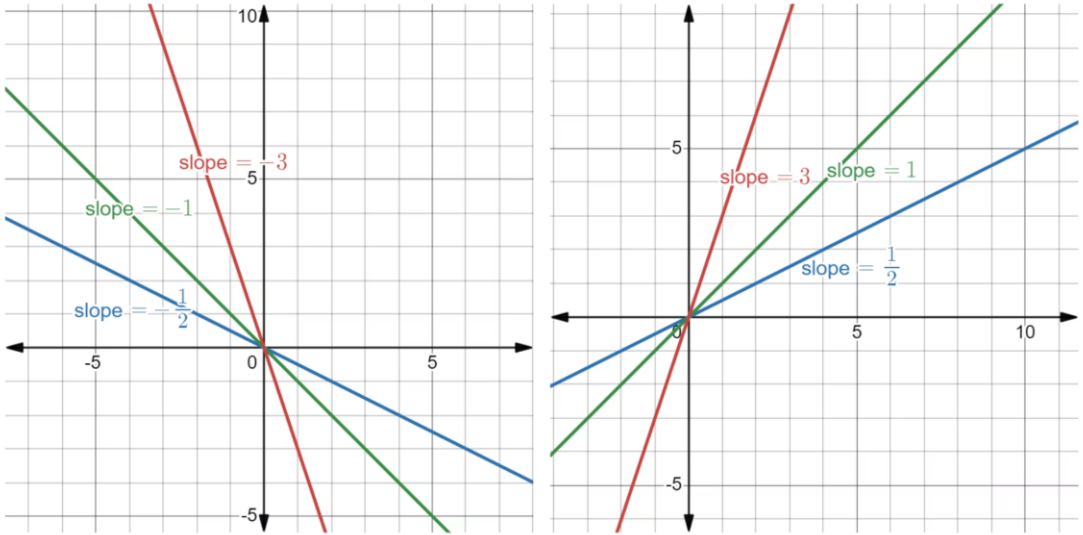
Some points fall below the line

The model over-predicted for these cases.

Goal: minimize errors overall

The best line keeps those gaps as small as possible across all points.

What Does the Coefficient Mean?



The slope β_1 tells us the expected change in Y for every one-unit increase in X.

$\beta_1 > 0$ (Positive)

Y tends to **increase** as X increases.
Example: more income → higher house price

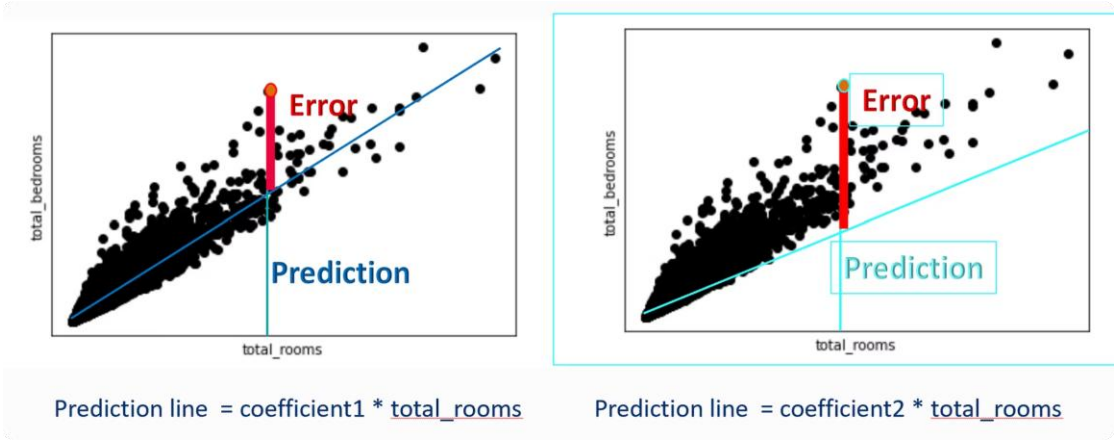
$\beta_1 < 0$ (Negative)

Y tends to **decrease** as X increases.
Example: more crime → lower house price

⚠ β_0 is the predicted Y when X = 0 — not always meaningful in real-world data!

How Do We Choose the Best Line?

Many lines can pass through the same data. Regression picks the one that **minimizes the total squared error** — this is called **Least Squares**.



Objective: Minimize $\Sigma(y_i - \hat{y}_i)^2$

Why square the errors?

Squaring ensures positive and negative errors don't cancel out.

Why not just absolute value?

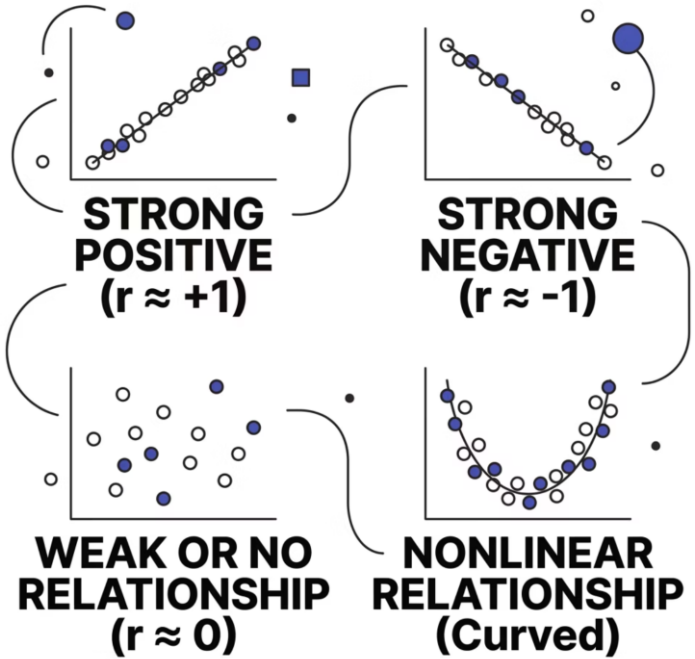
Squaring penalizes large errors more heavily — a useful property.

CLASS DISCUSSION

Why might we want large errors to be penalized more than small ones?

Pearson Correlation: Useful but Limited

Before fitting a model, use **Pearson's r** to quickly assess the linear relationship between two numerical variables.



r = +1

Perfect positive linear relationship

r = −1

Perfect negative linear relationship

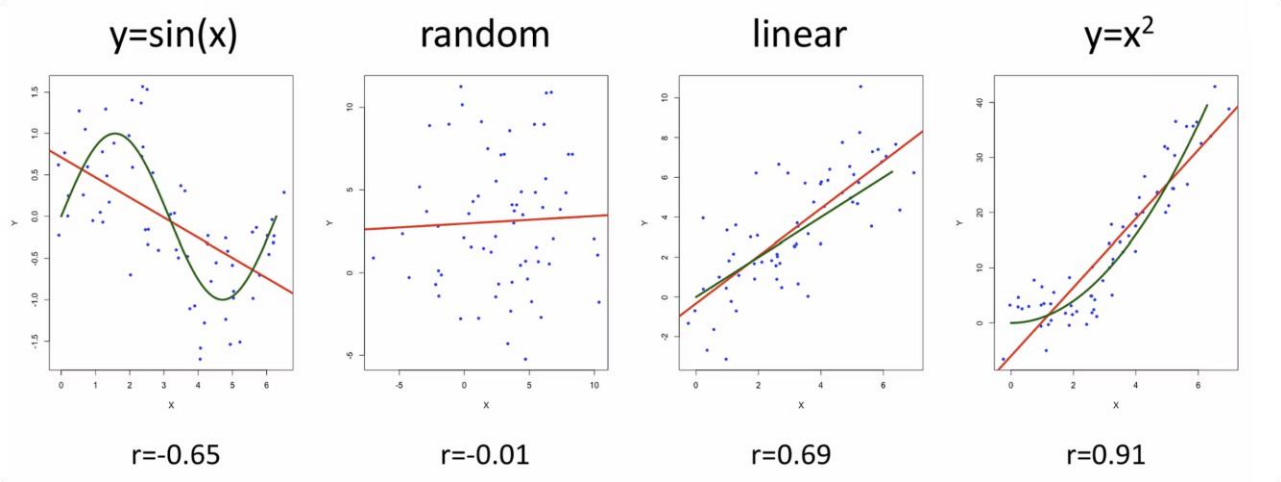
r ≈ 0

No linear relationship (may still have patterns!)

⚠ **Correlation ≠ Causation.** Always inspect the scatter plot — don't rely on r alone.

CLASS DISCUSSION

Correlation Doesn't Guarantee a Good Linear Model



A high r value can still hide **nonlinear behavior**. A low r can miss complex patterns. Always visualize your data before fitting a line.

✅ Linear

r = 0.69

High r, line fits well

❓ Random

r = -0.01

Low r, no clear structure

⚠ Sine/Wave

r = -0.65

Low r, but clear pattern

⚠ Quadratic

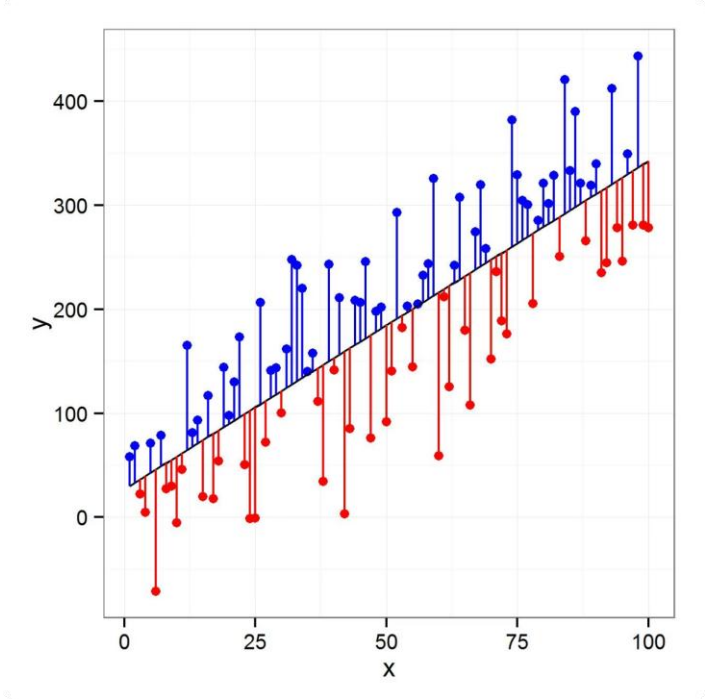
r = 0.91

Medium r, wrong model shape

Can a strong correlation prove causation? Give an example where it cannot.

Beyond Correlation: Assessing Linear Model Fit

Residuals: What the Model Didn't Explain



Residual formula: e = y − ŷ (observed minus predicted)

Small residuals = better fit

The predictions are close to reality.

Patterns in residuals = red flag

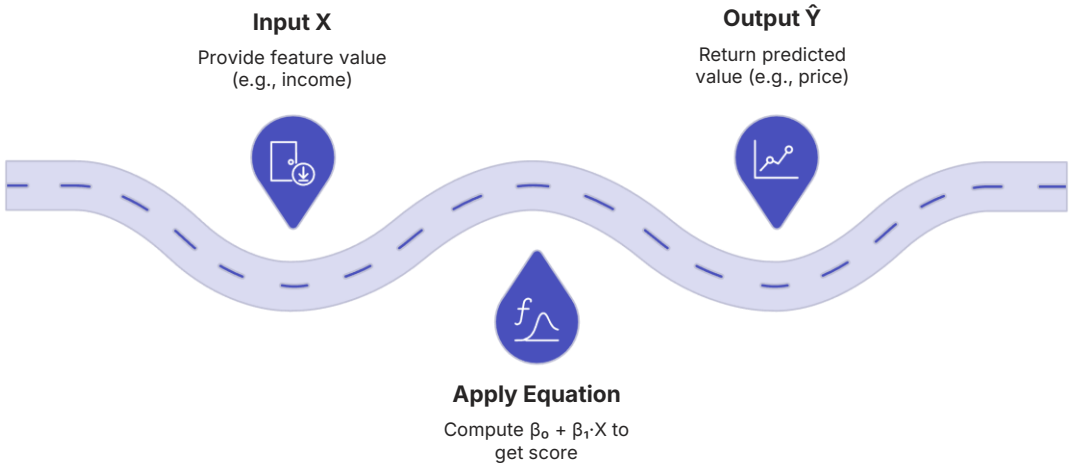
A curved residual plot suggests a linear model is the wrong shape.

Residuals are a diagnostic tool

Always plot them — don't just look at summary metrics.

Making Predictions with Simple Regression

The Prediction Flow



Once trained, the model is just an equation. Plug in X, and you instantly get a predicted Y.

Housing Example:
If β_0 = 45,000 and β_1 = 40,000:
A district with medianIncome = 5 → predicted house value = $245,000

Training Phase

Use existing data to estimate β_0 and β_1 via least squares.

Prediction Phase

Apply the equation to new X values to get $\hat{Y}$.

Implementing Linear Regression in Python

Python provides two primary libraries for building linear regression models: **scikit-learn** for machine learning tasks and **statsmodels** for statistical inference.

1. Scikit-learn: LinearRegression

Focus: Predictive modeling, integration into ML pipelines.

This model is optimized for making predictions and is easy to use within broader machine learning workflows.

```
from sklearn.linear_model import LinearRegression
import numpy as np

X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 5, 4, 5])

model = LinearRegression()
model.fit(X, y)

print(f"Intercept: {model.intercept_:.2f}")
print(f"Coefficient: {model.coef_[0]:.2f}")
print(f"Predictions for X=6: {model.predict([[6]])[0]:.2f}")
```

- Outputs primarily coefficients and intercept.
- Doesn't provide extensive statistical summaries.

2. Statsmodels: OLS

Focus: Statistical inference, detailed analysis of relationships.

OLS (Ordinary Least Squares) in statsmodels offers comprehensive statistical output for hypothesis testing and deeper insights.

```
import statsmodels.api as sm
import numpy as np

X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 5, 4, 5])

# Add a constant to the predictor (for intercept)
X_const = sm.add_constant(X)

model = sm.OLS(y, X_const)
results = model.fit()

print(results.summary())
```

- Provides p-values, R-squared, standard errors, etc.
- Ideal for research and understanding variable significance.

Key Differences: Prediction vs. Inference

1	2
Scikit-learn • Simple API for quick model building. • Focus on making accurate predictions. • Less emphasis on statistical detail.	**Statsmodels** • Rich statistical output and diagnostic tools. • Focus on understanding variable relationships. • More academic and research-oriented.

Regression Evaluation Metrics

How do we measure how well our regression model performs? Use these four metrics together.

1

MSE

Mean Squared Error — average of squared errors. Penalizes large mistakes heavily.

2

MAE

Mean Absolute Error — average of absolute errors. Intuitive and in original units (e.g., dollars).

3

RMSE

Root MSE — square root of MSE. Also in original units, easier to compare to MAE.

4

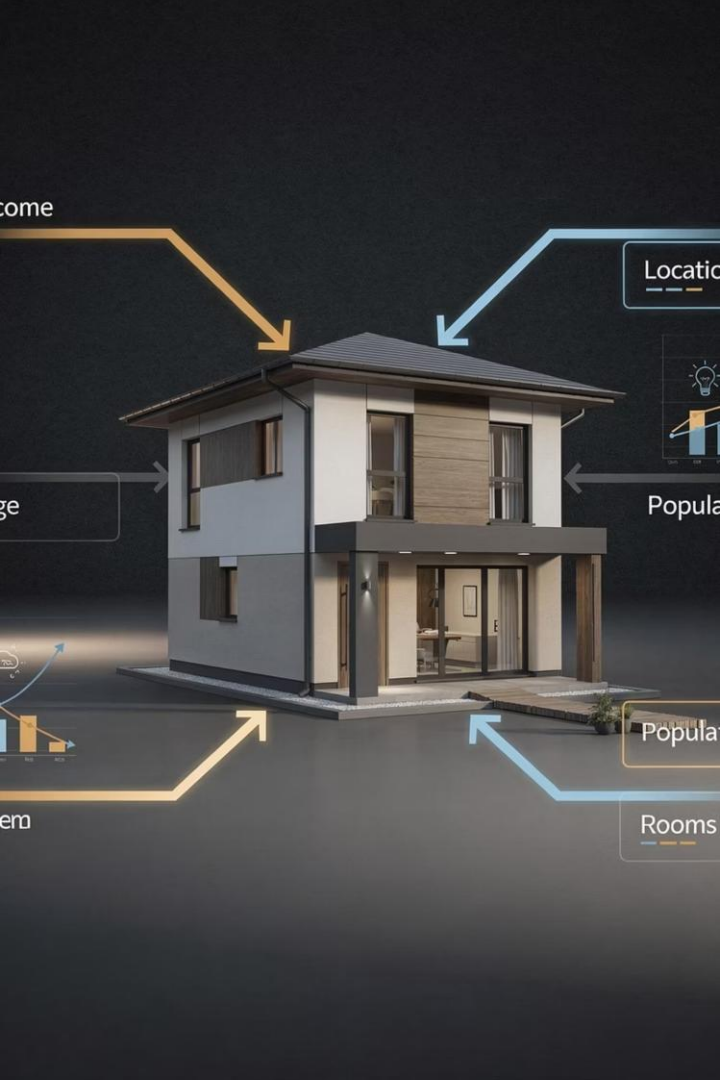
R^2

Variance explained. Ranges 0–1 on training data; can be **negative on test data** if model is worse than predicting the mean.

For house prices, **MAE is especially interpretable** — it tells you the average error in dollars.

CLASS DISCUSSION

Why might a model have low MSE but still be hard to interpret? When would MAE be preferred over RMSE?



Why One Predictor Is Often Not Enough

House prices are shaped by **many factors simultaneously**. A single-variable model leaves out crucial information.

Median Income
Strong predictor of house values.

Housing Median Age
Older neighborhoods may command premiums.

Total Rooms
More rooms often mean higher prices.

Total Bedrooms
Reflects capacity and family suitability.

Population
Signals density and area demand.

Households
Shows residential community size.

Latitude
Strongly shapes California home prices.

Longitude
Combines with latitude to locate homes.

Ocean Proximity
Ocean access adds major premium.

Multiple Linear Regression

The same idea as simple regression — but now we use **many predictors at once**, each with its own coefficient.

Formula:
$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p$

β_0

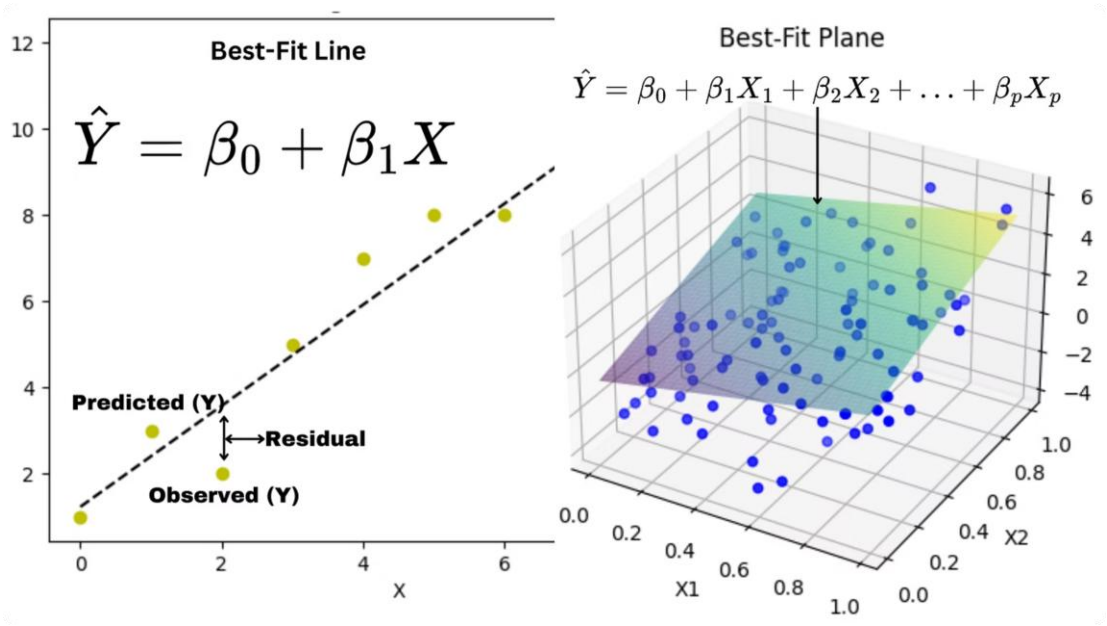
Intercept — baseline prediction

$\beta_1 \ldots \beta_p$

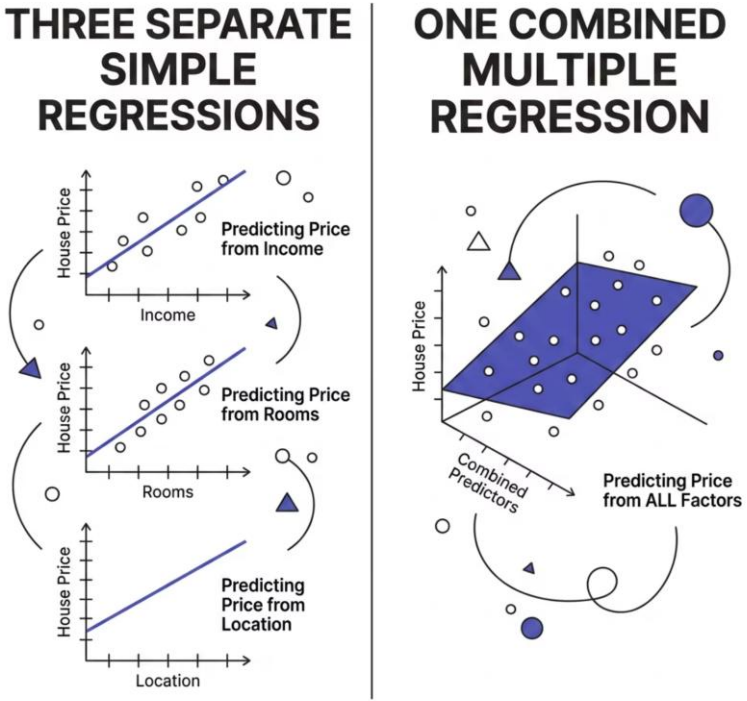
Each predictor's unique contribution to $\hat{Y}$

One model, one prediction

All variables are combined into a single output value



Why Not Build Separate Simple Models?



No combined output

Three separate models cannot produce one unified prediction.

Ignores variable interactions

Predictors are correlated — treating them separately distorts coefficients.

Misleading importance

A variable may look important alone but become redundant once others are included.

CLASS DISCUSSION

What happens to model accuracy if we add irrelevant or redundant variables?

Preparing Housing Data for Multiple Regression

Regression models require **clean, numerical inputs**. Here's the preprocessing pipeline for the California Housing dataset.



Numerical Features

longitude, latitude, housing_median_age, total_rooms, total_bedrooms, population, households, median_income

Categorical Feature

ocean_proximity → one-hot encoded into binary columns

Target Variable

medianHouseValue — the number we want to predict

Interpreting Multiple Regression Coefficients

longitude	-267,834
latitude	-241,440
housing_median_age	55,150
total_rooms	248,574
total_bedrooms	663,758
population	-1,299,063
households	262,351
median_income	569,525
ocean_proximity_<1H OCEAN	-34,269
ocean_proximity_INLAND	-73,509
ocean_proximity_ISLAND	179,383
ocean_proximity_NEAR BAY	-40,501
ocean_proximity_NEAR OCEAN	-31,102

Each coefficient describes the predicted change in house value **holding all other variables fixed**.

⚠ **Scale matters!** Coefficients are not directly comparable unless features are standardized. A large coefficient may just reflect small-scale units.

⚠ ⚠ Without standardization, coefficients reflect raw units — not feature importance. Large values (e.g. population) may just reflect small-scale units.

Remember: coefficients describe **associations**, not causal effects.

Positive coefficient

Higher median income → higher predicted price. Ocean proximity → premium value.

Negative coefficient

Higher population density may predict lower prices in some models.

🔍 KEY INSIGHT

Regression Coefficients After Scaling

After standardizing features (z-score scaling), coefficients become comparable — revealing true feature importance.

population: -2.678

Strongest negative driver. Higher population → lower predicted price.

total_bedrooms: +1.369

More bedrooms → higher predicted price.

median_income: +1.174

Higher income neighborhood → higher predicted price.

longitude: -0.561

Further west → slightly lower predicted price.

ⓘ 📈 Prices are higher as total_bedrooms and median_income of the neighborhood are higher

⚠ 📉 Prices are lower as population is higher and longitude is higher — or higher as population is lower and longitude is lower

Model Evaluation: Multiple Regression Results

Mean Squared Error (MSE): 4,802,173,538.60

Formula: average of (actual − predicted)²

Very sensitive to large errors due to squaring. Hard to interpret directly because of large scale.

R^2 (Coefficient of Determination): 0.649

Proportion of variance in house prices explained by the model.

Interpretation: ~65% of price variation is captured by our features.

Mean Absolute Error (MAE): $50,413

Average absolute difference between predicted and actual prices.

Most interpretable: on average, our predictions are off by $50,413.

🏠 Practical meaning: We can predict a California house price with a mean error of ~$50,413 — roughly 25% of the median house value.

📊 MULTIPLE REGRESSION ON CALIFORNIA HOUSING

Going Deeper: MAPE & R^2 Benchmarks

Why MAPE May Be Better

MAPE (Mean Absolute Percentage Error)

Expresses error as a percentage of the actual value. Formula: average of |actual − predicted| / actual × 100. Example: MAPE of 18% means predictions are off by 18% on average — easy to communicate to non-technical stakeholders.

Why prefer MAPE?

Scale-independent — works across datasets with different price ranges. Easier to interpret than MSE or MAE alone. Caveat: unstable when actual values are near zero.

What Are Normal R^2 Values?

R^2 Range	Interpretation
< 0.3	Weak fit — model explains little variance
0.3 – 0.7	Moderate fit — acceptable for complex real-world data
> 0.7	Strong fit — model captures most variance

Our model: R^2 = 0.649 → moderate fit, typical for housing data with many unobserved factors.

How to Improve Your Model & Handle Outliers

How to Improve R^2

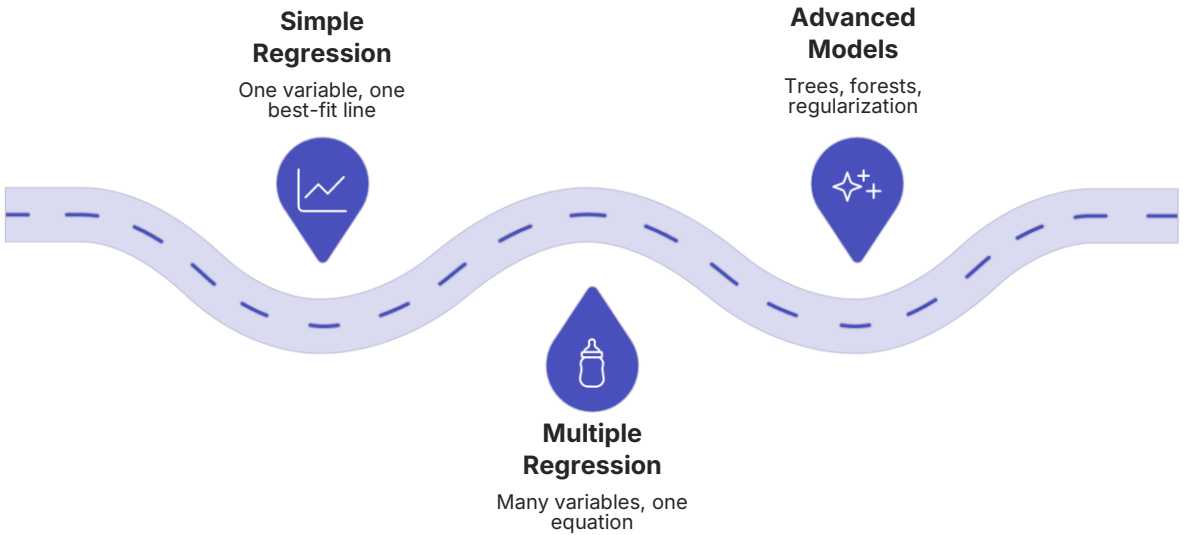
- **Add more relevant features**
  e.g. school quality, crime rate, proximity to amenities
- **Engineer new features**
  e.g. rooms_per_household, bedrooms_ratio
- **Try non-linear models**
  Decision Trees, Random Forests, Gradient Boosting
- **Remove or handle outliers that distort the fit**

The Role of Outliers

⚠ **⚠ Outliers have an outsized effect on linear regression because MSE squares the errors — a single extreme point can pull the regression line significantly. Always visualize residuals and check for high-leverage points.**

Detect outliers	Handle outliers
Use residual plots, box plots, or z-score thresholds ($\|z\| > 3$)	Remove if data entry errors; cap/transform if genuine extreme values; or use robust regression methods

Summary & Key Takeaways



01

Regression predicts continuous numbers

Not classes — the output is a value like house price or income.

02

Least squares finds the best-fit line

By minimizing the sum of squared residuals.

03

Residuals and metrics evaluate fit

Use MSE, MAE, RMSE, and R^2 together — never just one.

04

Multiple regression combines all predictors

One model, one prediction, many inputs — more powerful than separate models.

Coming up next: Can we do even better? → Decision Trees, Random Forests, Gradient Boosting, and Regularization (Ridge / Lasso).

Thank You!

Go explore data — get your hands dirty, break things, and learn by doing.

Use chatbots to help you code and explore — but guide them intelligently and always judge their results critically.

Email

MarcelFei@telhai.ac.il

Office Hours

Per Request

Location

Building 6, Room 309, Eastern Campus